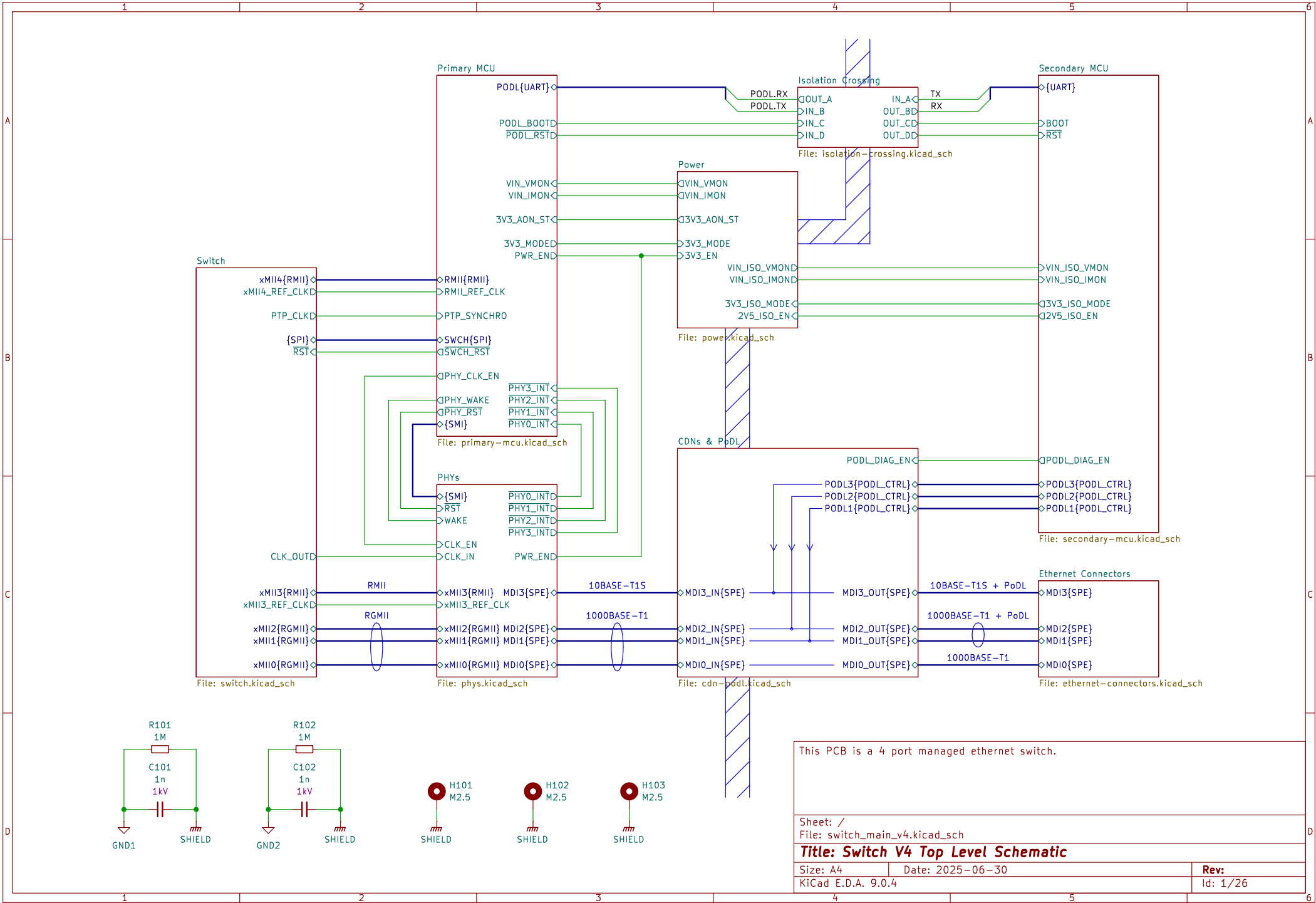




This PCB is a 4 port managed ethernet switch.

Sheet: /
File: switch_main_v4.kicad_sch

Title: Switch V4 Top Level Schematic

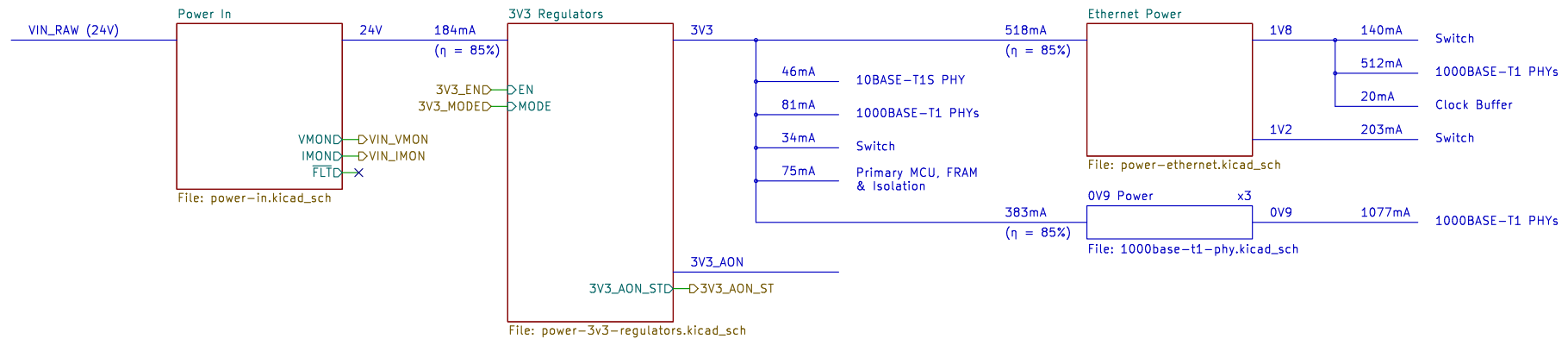
Size: A4 Date: 2025-06-30

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Rev:

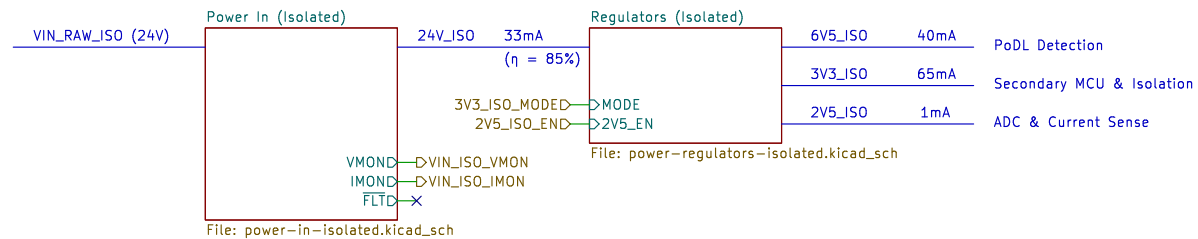
Id: 1/26

PHY Side Power



Isolation Barrier

Connector Side Power



Block diagram of the power supply scheme for both sides of the isolation barrier.

Sheet: /Power/
File: power.kicad_sch

Title: Power

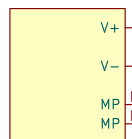
Size: A4	Date: 2025-06-30
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Rev:

Id: 2/26

J301
MKXS1-1 Power Input



ESD
Protection

Surge
Protection

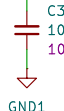
VIN_RAW

D301
SZESD7241N2T5G



U301
TVS3301DRBR

GND1



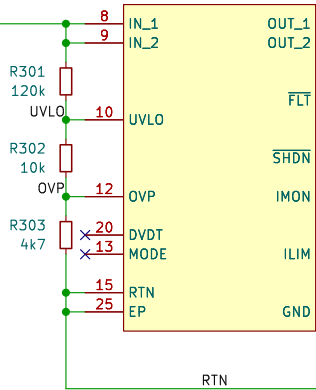
C301
100n
100V

GND1

UVLO = 10.9V
OVP = 34.1V

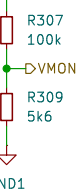
eFuse

U302
TPS26600RHFR



VMON = VIN / 18.857
The 150mΩ eFuse
resistance should be
taken into account when
calculating input power.

+VIN



I_{OUT} = V_{IMON} / 7.84

R_{ILIM} = 47kΩ ∴
I_{LIM} = 285mA

TPS26600 has auto-retry
when a fault occurs.

Power input and protection for the PHY side of the PCB.

Sheet: /Power/Power In/
File: power-in.kicad_sch

Title: Power In (PHY Side)

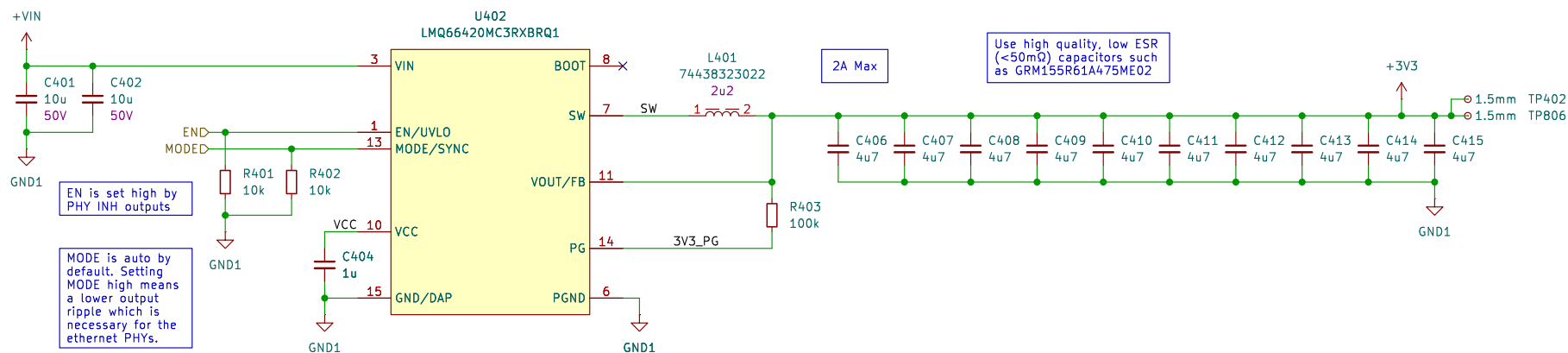
Size: A4 Date: 2025-06-30

KiCad E.D.A. 9.0.4

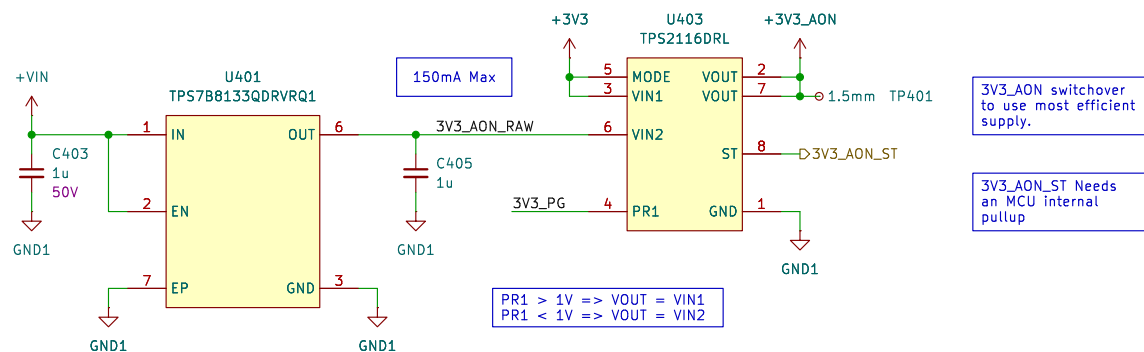
Rev:

Id: 3/26

3V3 Main Regulator



3V3 Always on supply



3V3 and 3V3_AON (always on) voltage regulators.

Sheet: /Power/3V3 Regulators/
File: power-3v3-regulators.kicad_sch

Title: 3V3 Regulators (PHY Side)

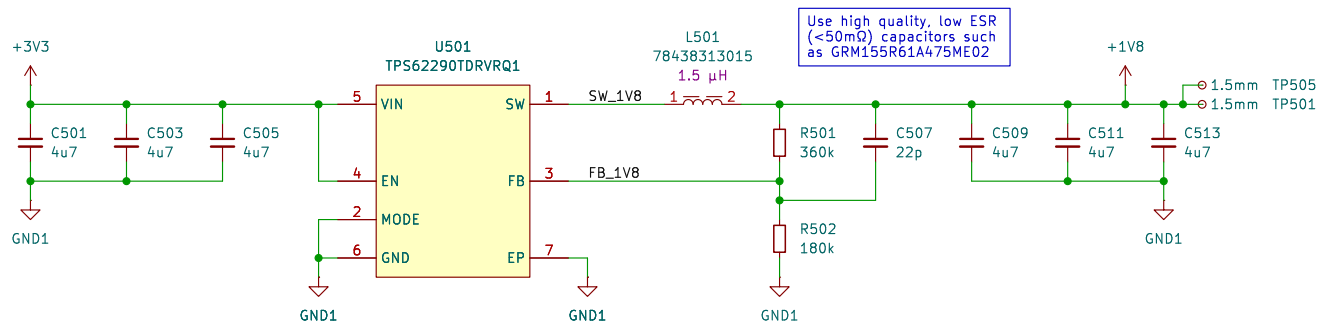
Size: A4 Date: 2025-06-30

KiCad E.D.A. 9.0.4

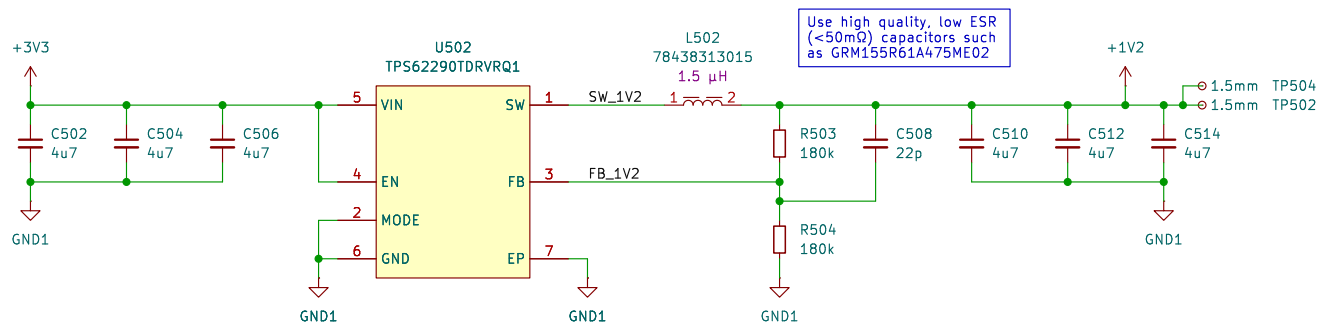
Rev:

Id: 4/26

1V8 Regulator



1V2 Regulator



1V8 and 1V2 SMPS for ethernet switch and PHYs.

Sheet: /Power/Ethernet Power/
File: power-ethernet.kicad_sch

Title: Ethernet Power (PHY Side)

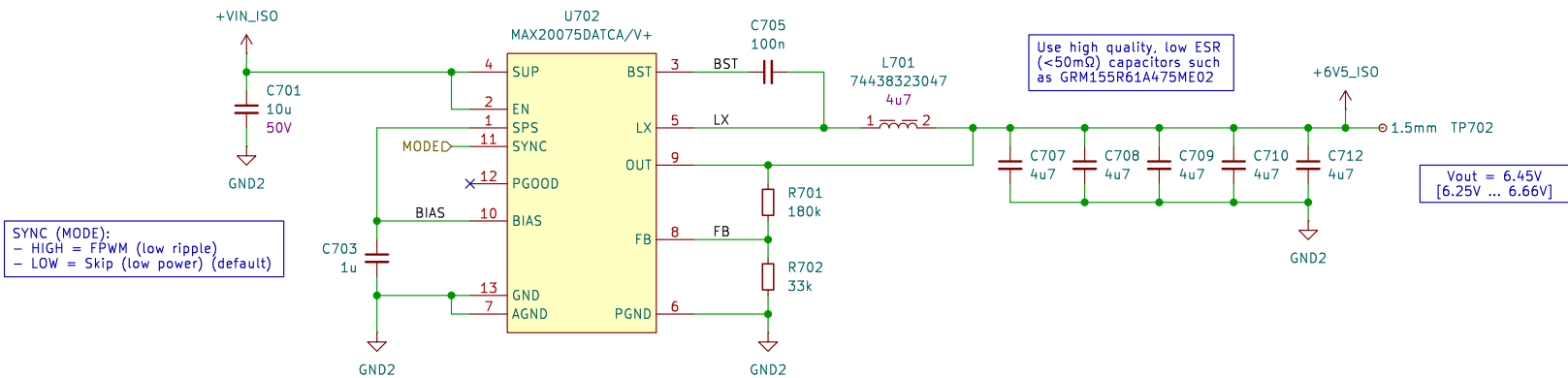
Size: A4 Date: 2025-06-30

KiCad E.D.A. 9.0.4

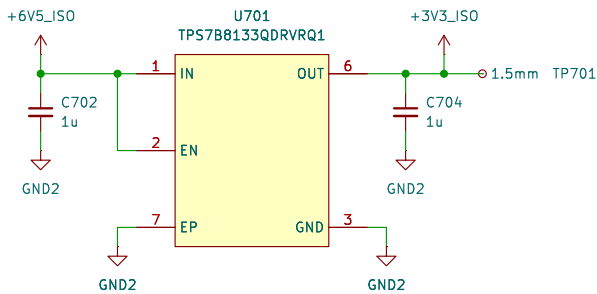
Rev:

Id: 5/26

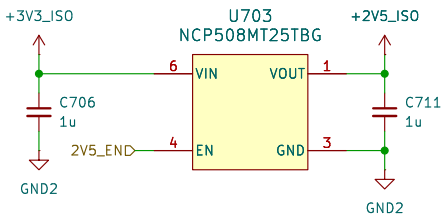
6V5 Regulator



3V3 Regulator



2V5 Regulator



6V5, 3V3 and 2V5 regulators for the connector side of the PCB.

Sheet: /Power/Regulators (Isolated)/
File: power-regulators-isolated.kicad_sch

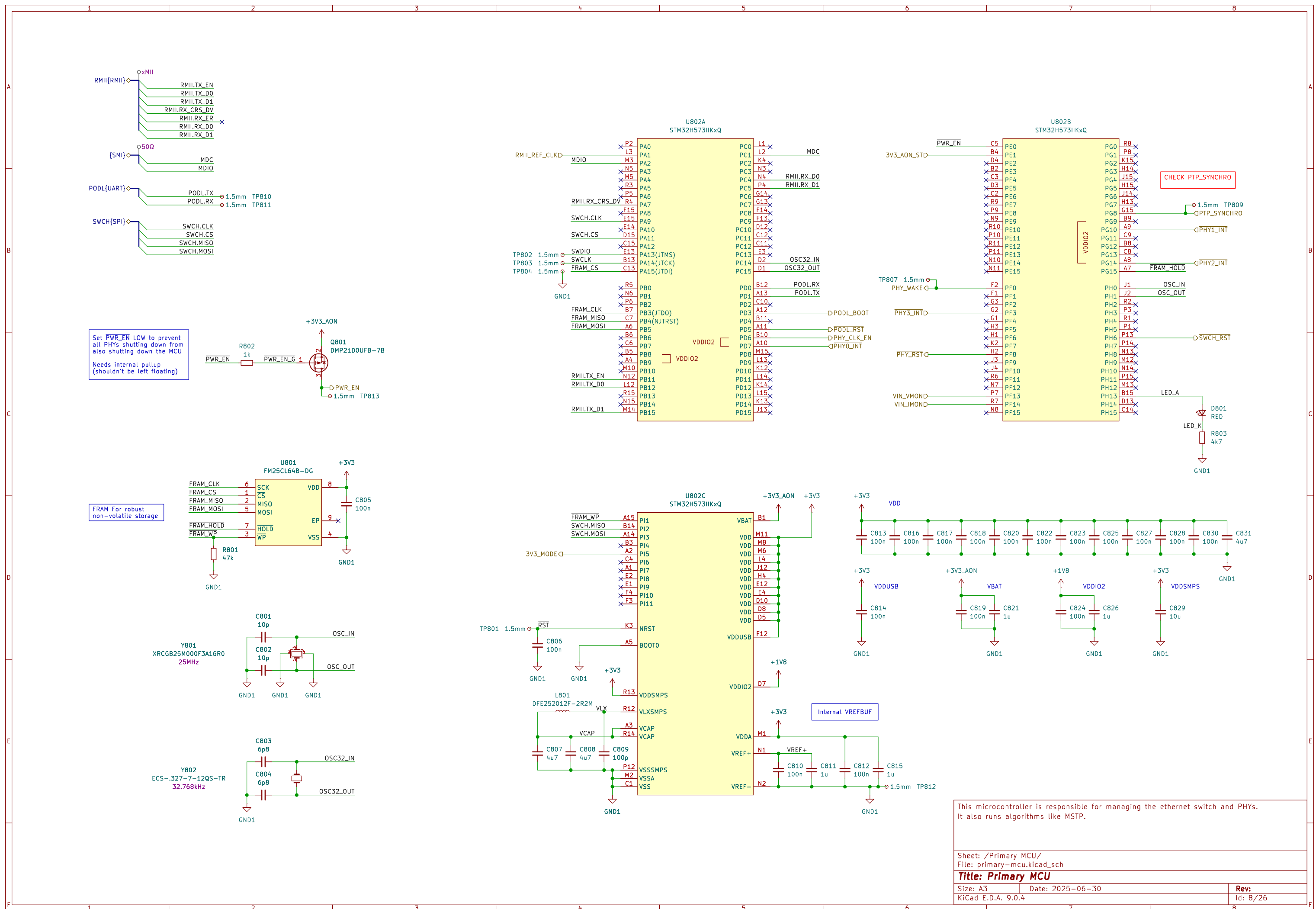
Title: Regulators (Connector Side)

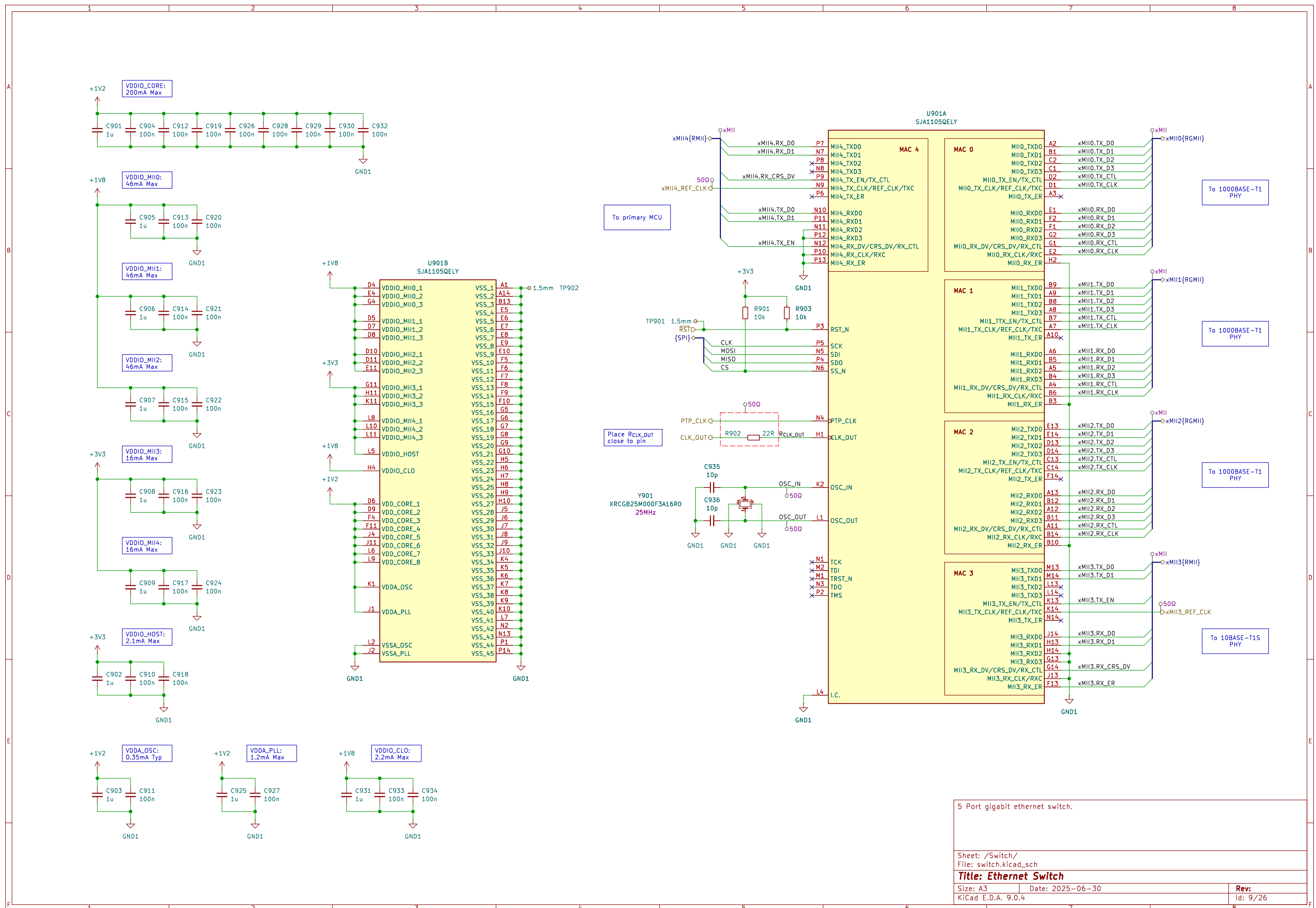
Size: A4 Date: 2025-06-30

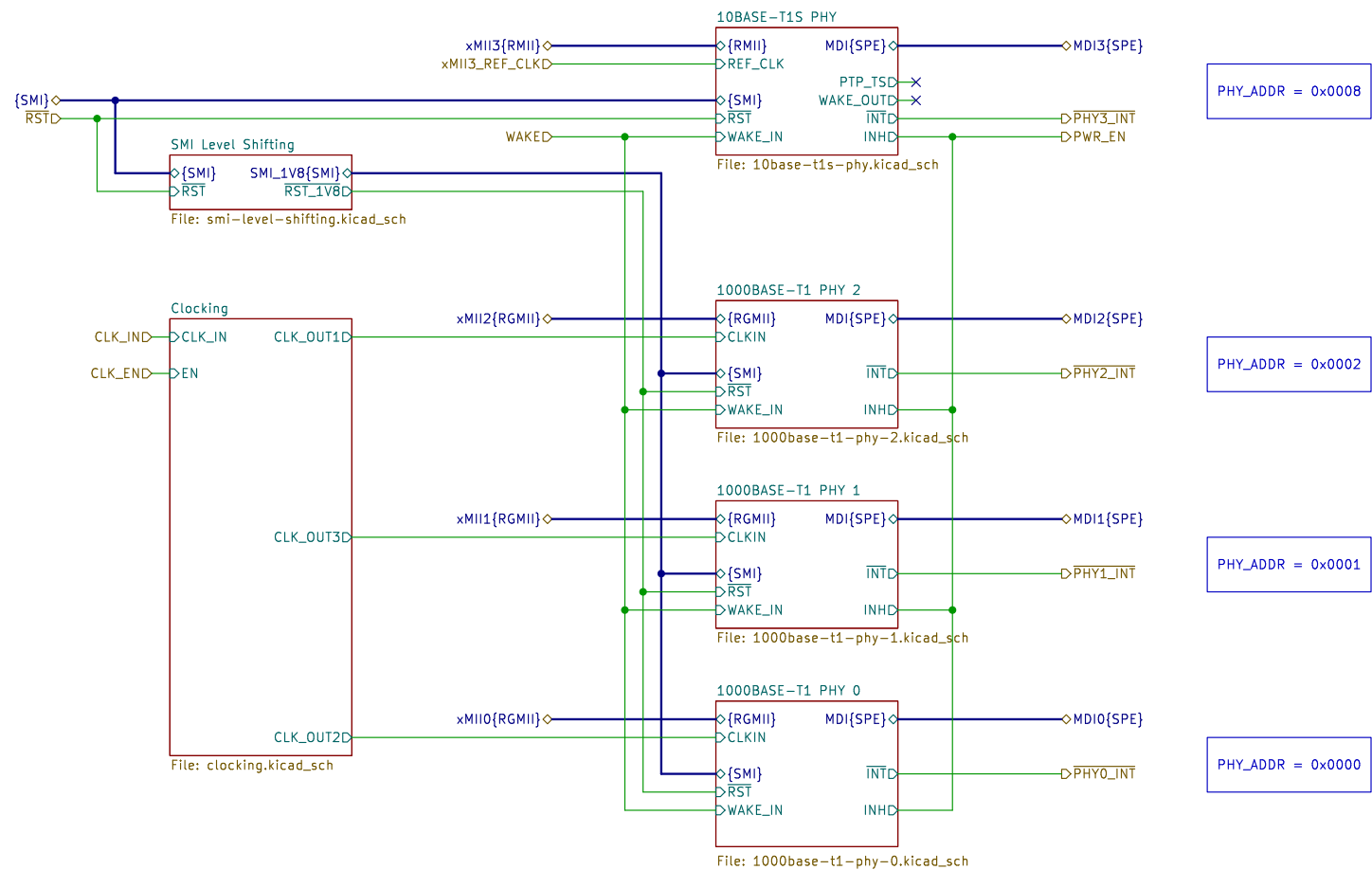
KiCad E.D.A. 9.0.4

Rev:

Id: 7/26







Ethernet PHYs and supporting circuitry.

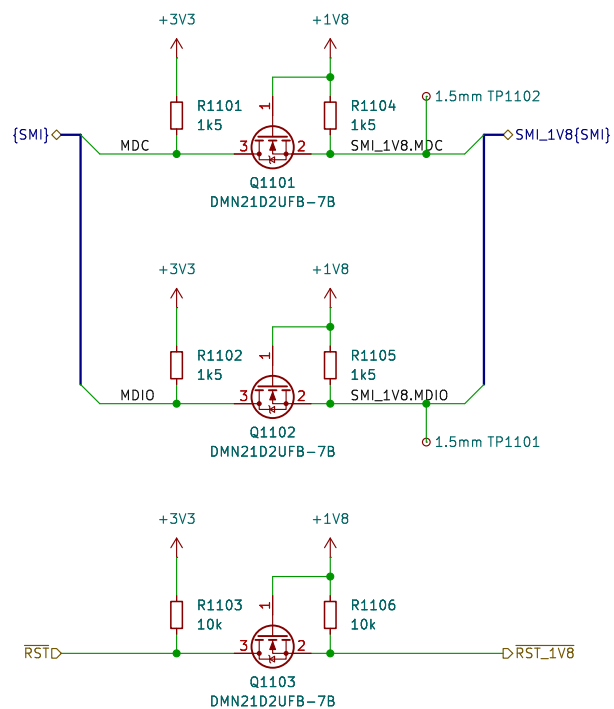
Sheet: /PHYs/
File: phys.kicad_sch

Title: PHYs

Size: A4 Date: 2025-06-30

KiCad E.D.A. 9.0.4

Rev:
Id: 10/26



The 1000BASE-T1 PHYs operate at a 1V8 logic level (to save RGMII power), but the SMI and RST pins operate at a 3V3 logic level. Level shifting is used to allow communication.

Converts 3.3V PHY management signals to 1.8V.

Sheet: /PHYs/SMI Level Shifting/
File: smi-level-shifting.kicad_sch

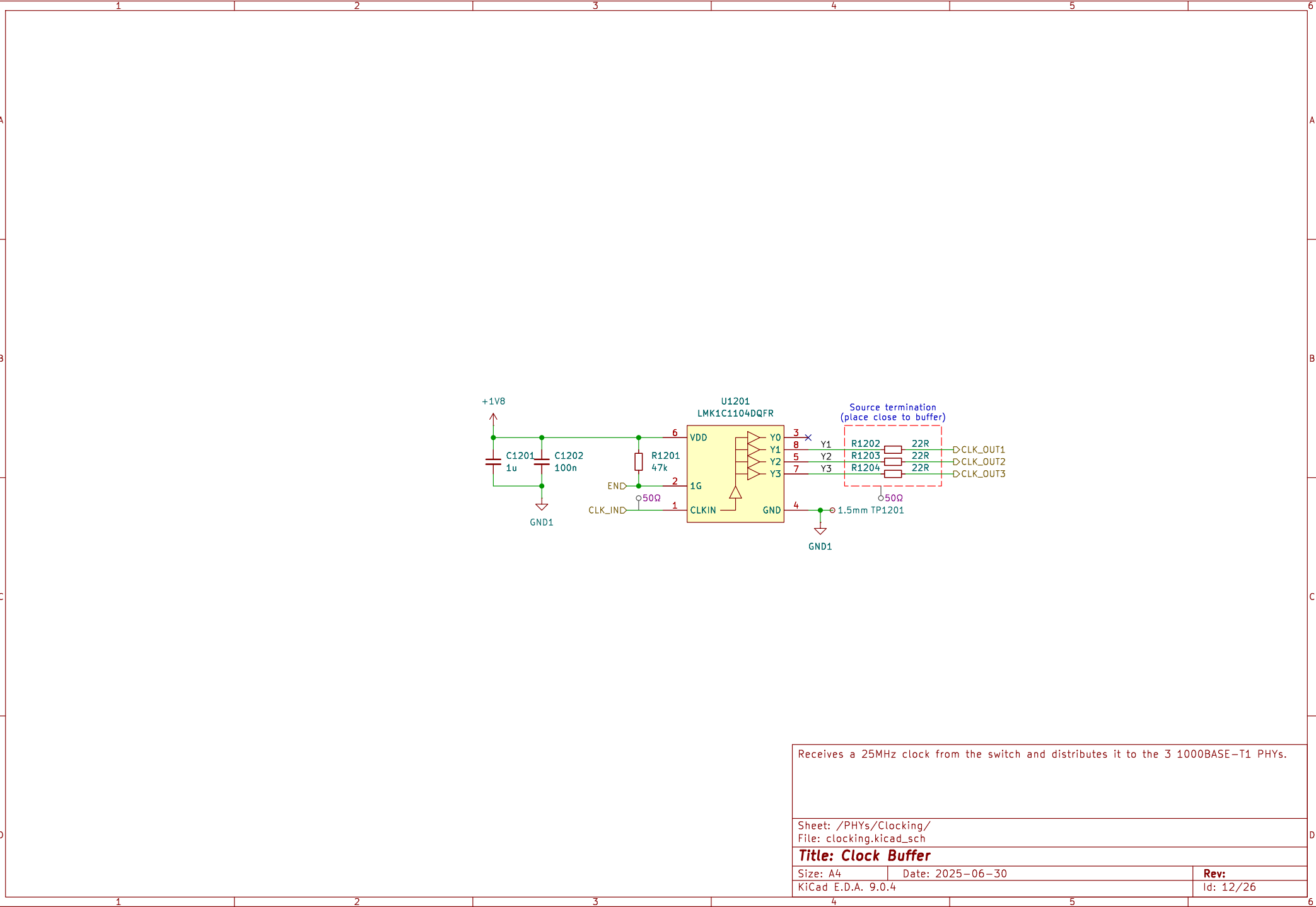
Title: SMI Level Shifting

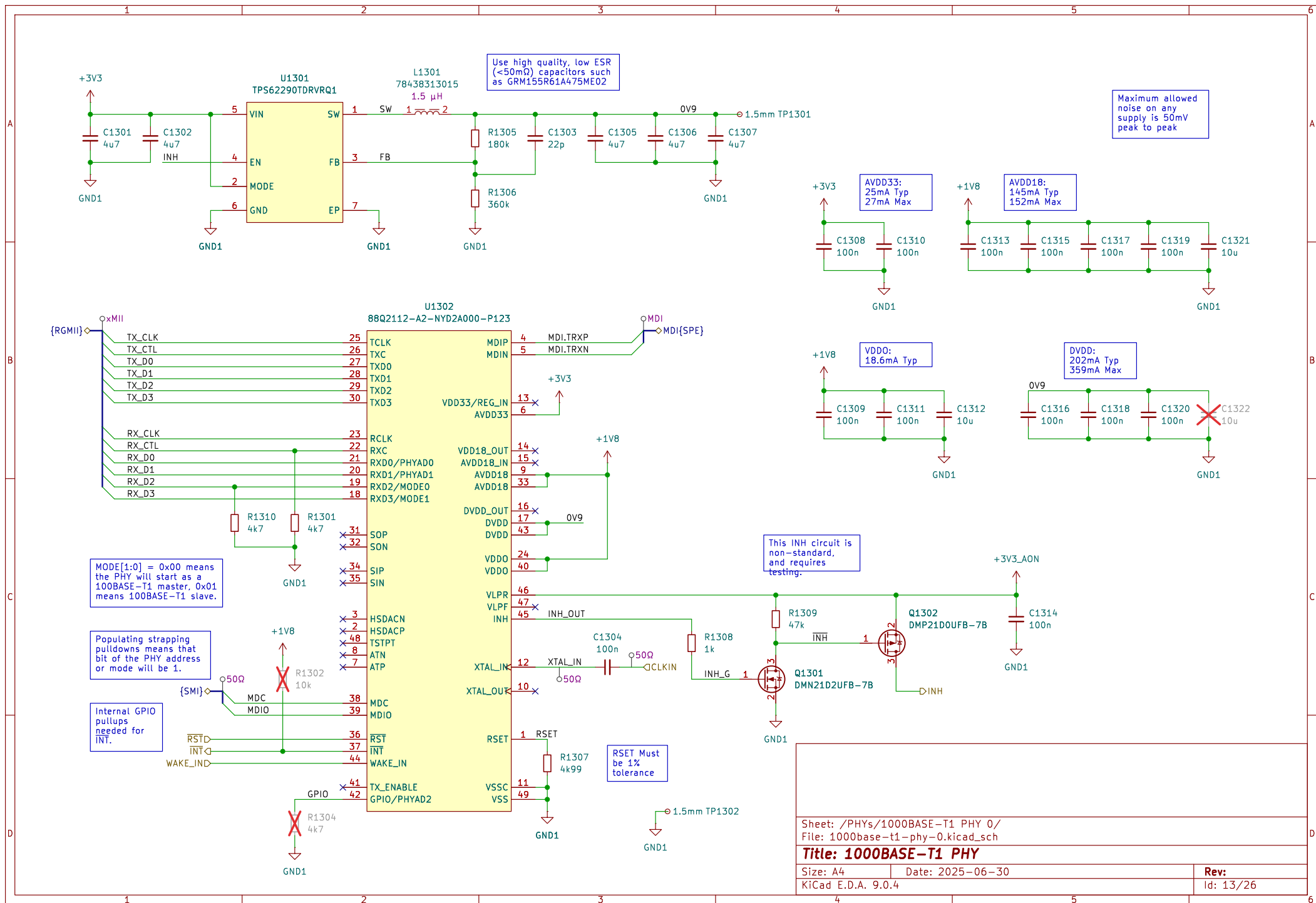
Size: A4 Date: 2025-06-30

KiCad E.D.A. 9.0.4

Rev:

Id: 11/26





Sheet: /PHYs/1000BASE-T1 PHY 0/
File: 1000base-t1-phy-0.kicad_sch

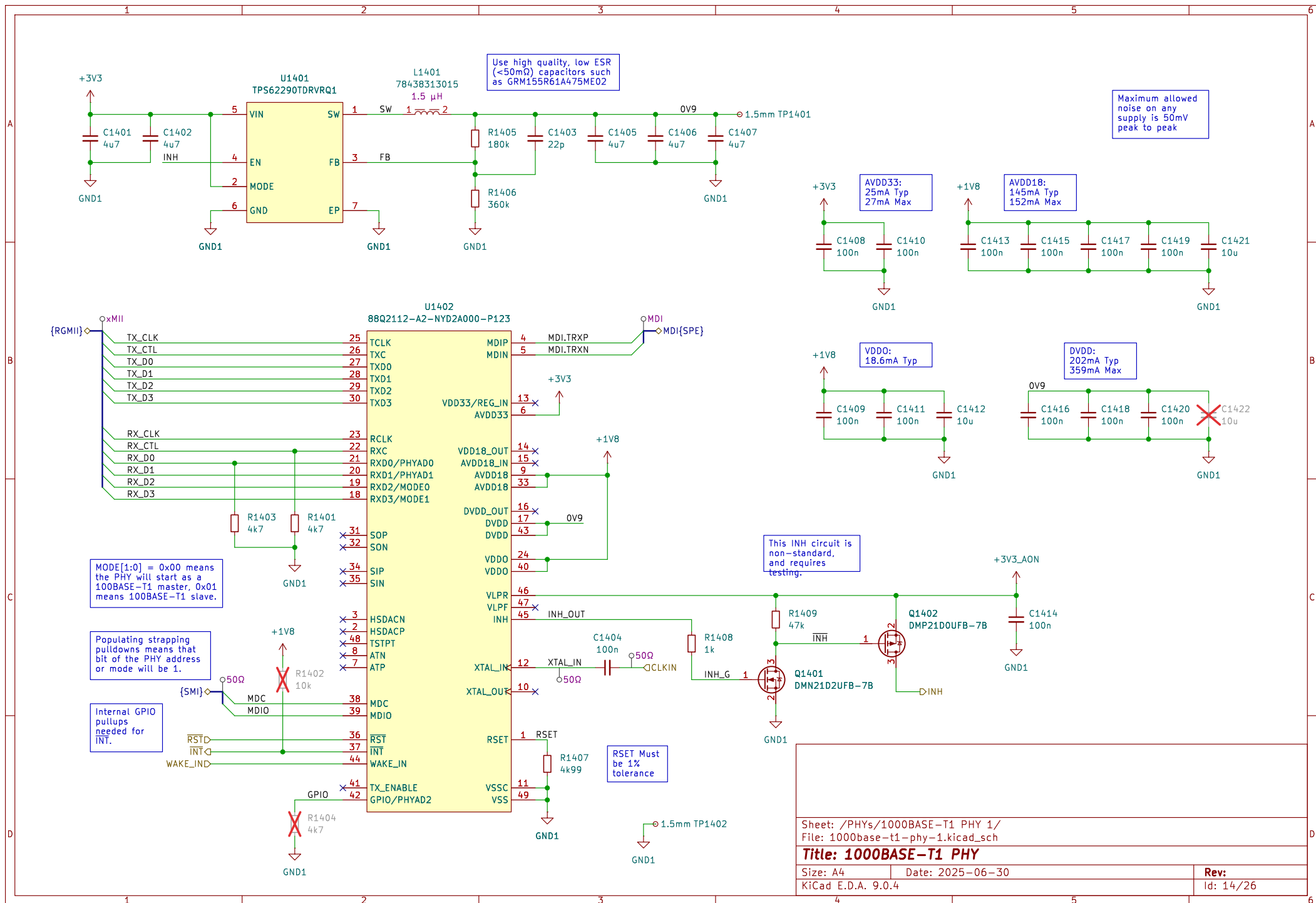
Title: 1000BASE-T1 PHY

Size: A4 Date: 2025-06-30

KiCad E.D.A. 9.0.4

Rev:

Id: 13/26



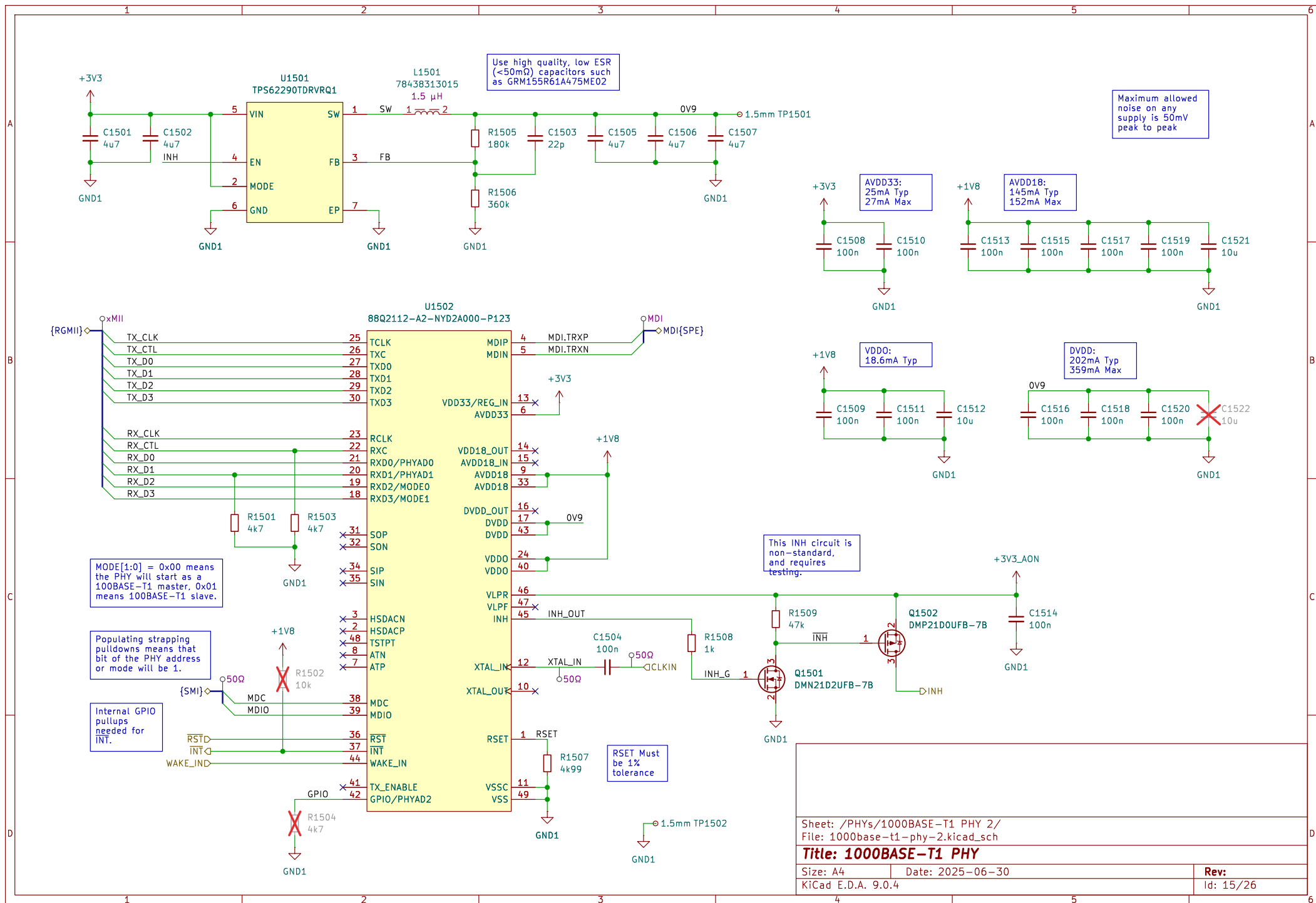
Sheet: /PHYs/1000BASE-T1 PHY 1/
File: 1000base-t1-phy-1.kicad_sch

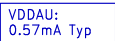
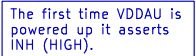
Title: 1000BASE-T1 PHY

Size: A4 Date: 2025-06-30

KiCad E.D.A. 9.0.4

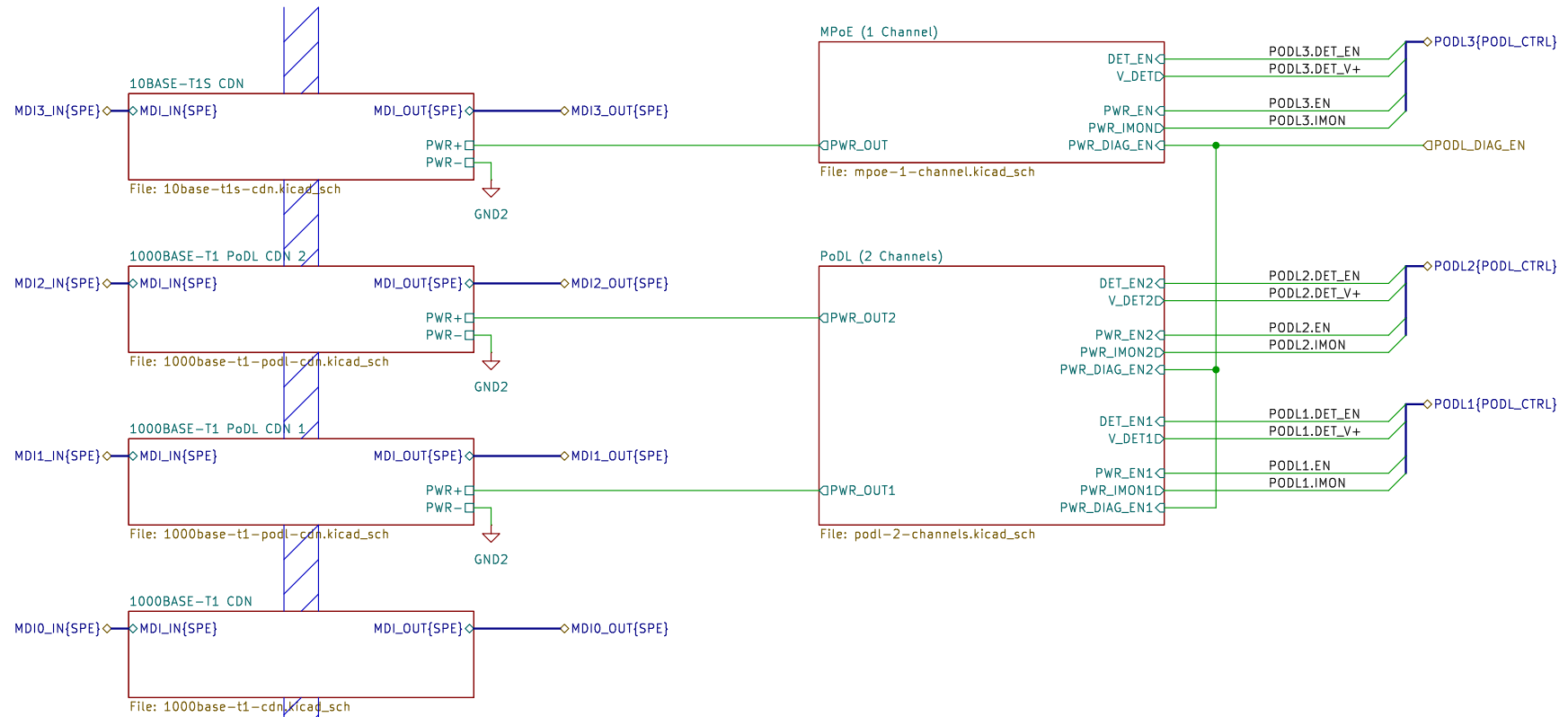
Rev:
Id: 14/26





Coupling-Decoupling Networks (CDNs)

Power Injection



Coupling-decoupling networks (CDNs) and power over datalines (PoDL).

Sheet: /CDNs & PoDL/
File: cdn-podl.kicad_sch

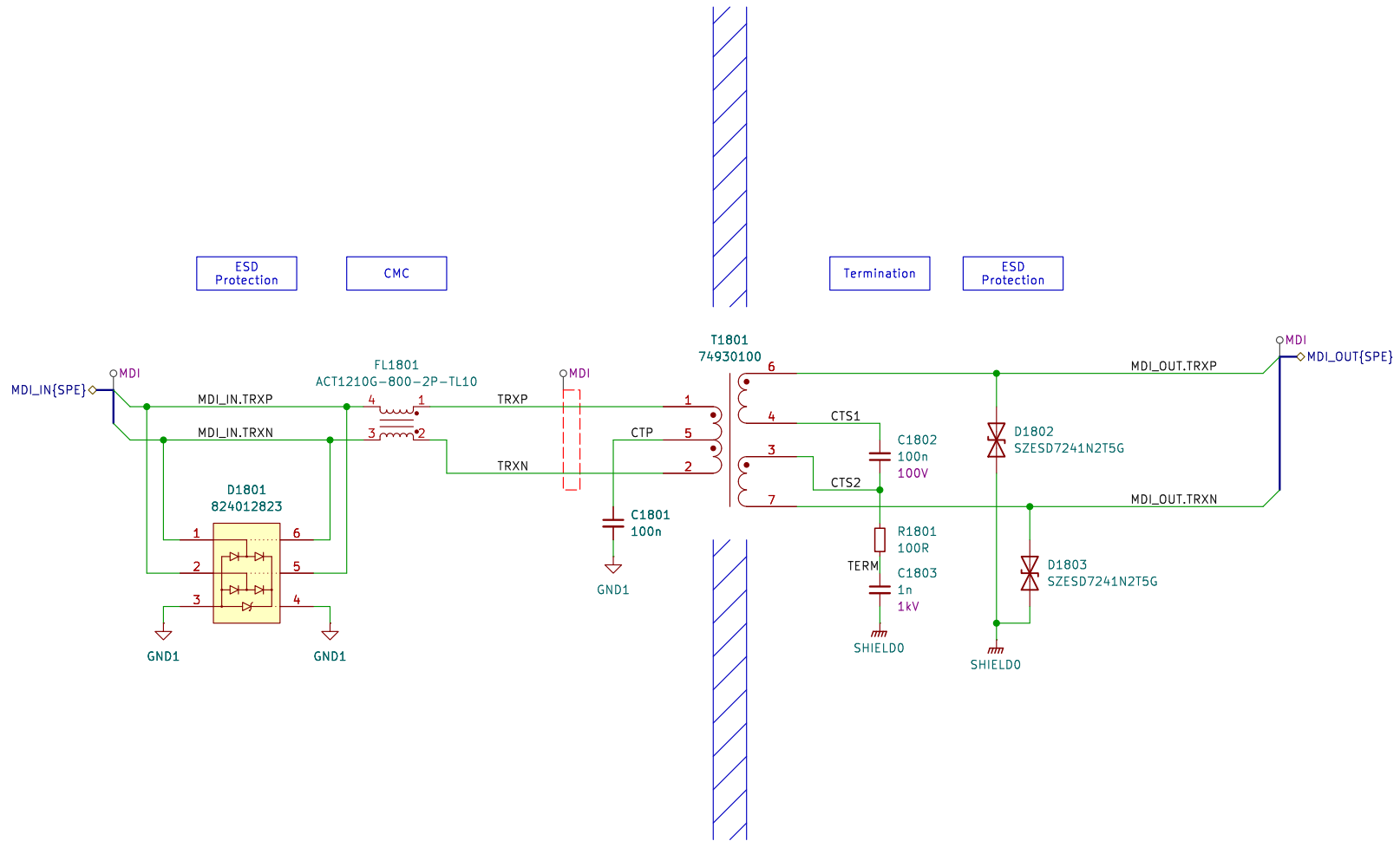
Title: CDNs and PoDL

Size: A4 Date: 2025-06-30

KiCad E.D.A. 9.0.4

Rev:

Id: 17/26



Coupling-decoupling network for 1000BASE-T1

Sheet: /CDNs & PoDL/1000BASE-T1 CDN/
File: 1000base-t1-cdn.kicad_sch

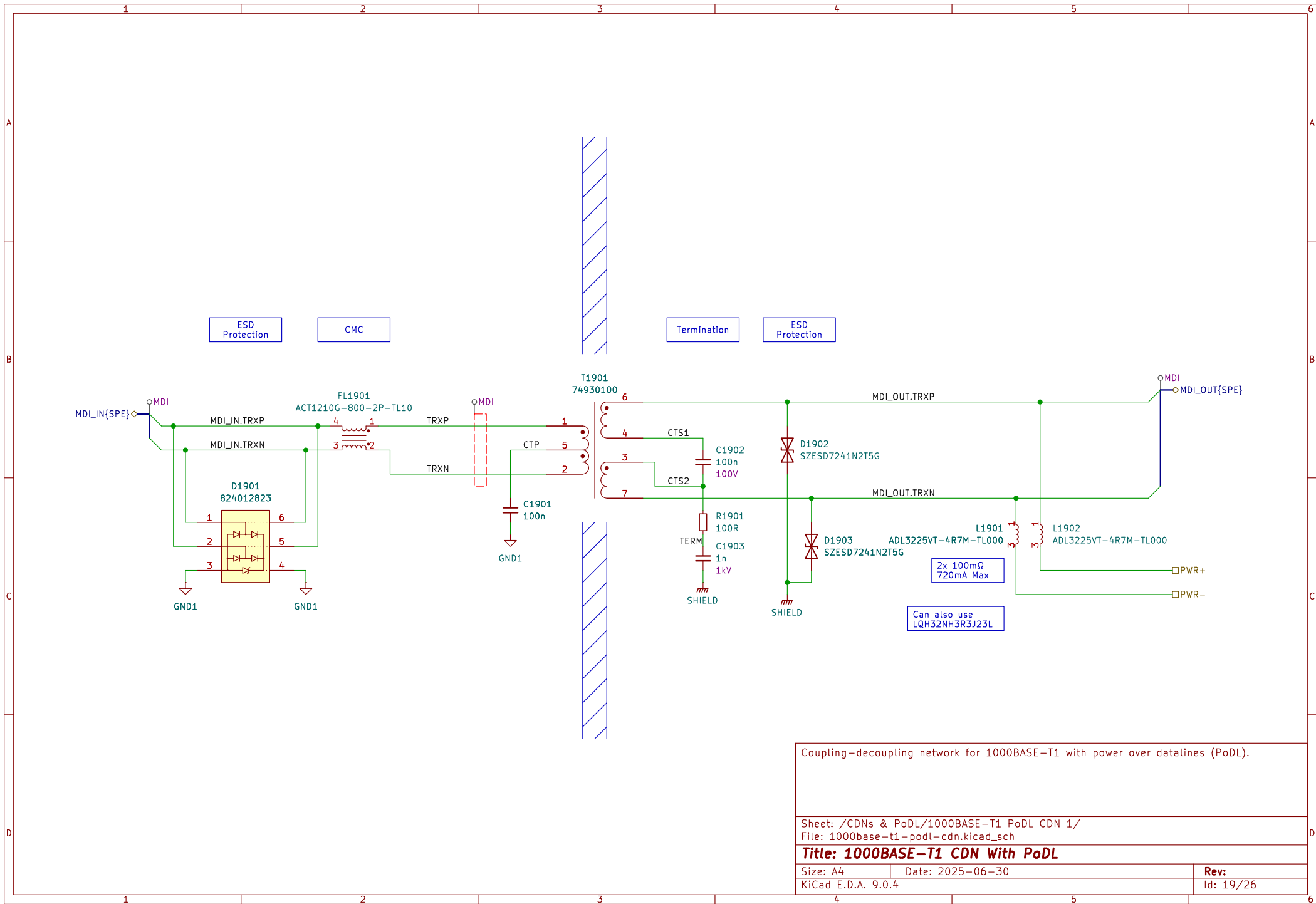
Title: 1000BASE-T1 CDN

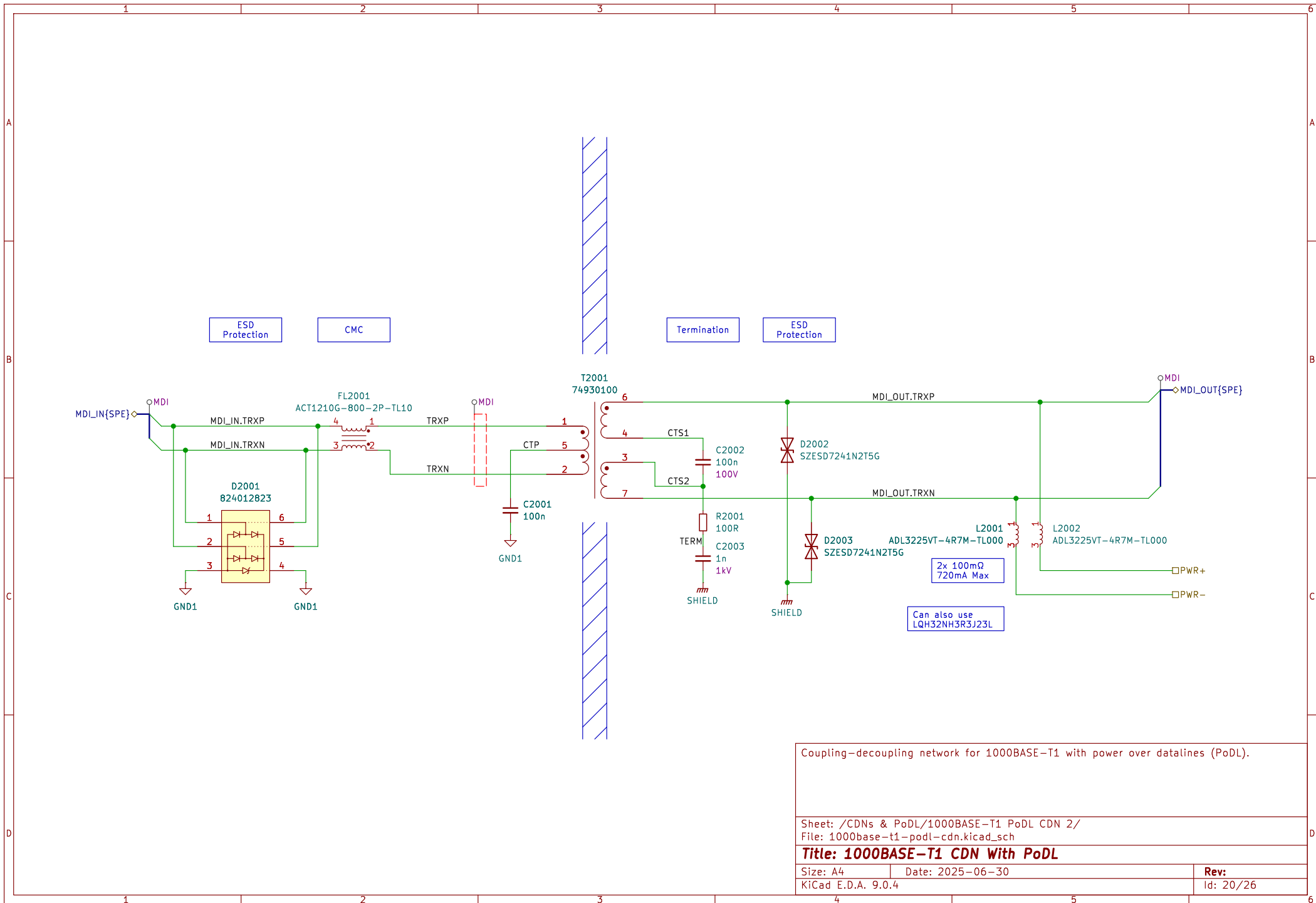
Size: A4 Date: 2025-06-30

KiCad E.D.A. 9.0.4

Rev:

Id: 18/26





Coupling-decoupling network for 1000BASE-T1 with power over datalines (PoDL).

Sheet: /CDNs & PoDL/1000BASE-T1 PoDL CDN 2/

File: 1000base-t1-podl-cdn.kicad_sch

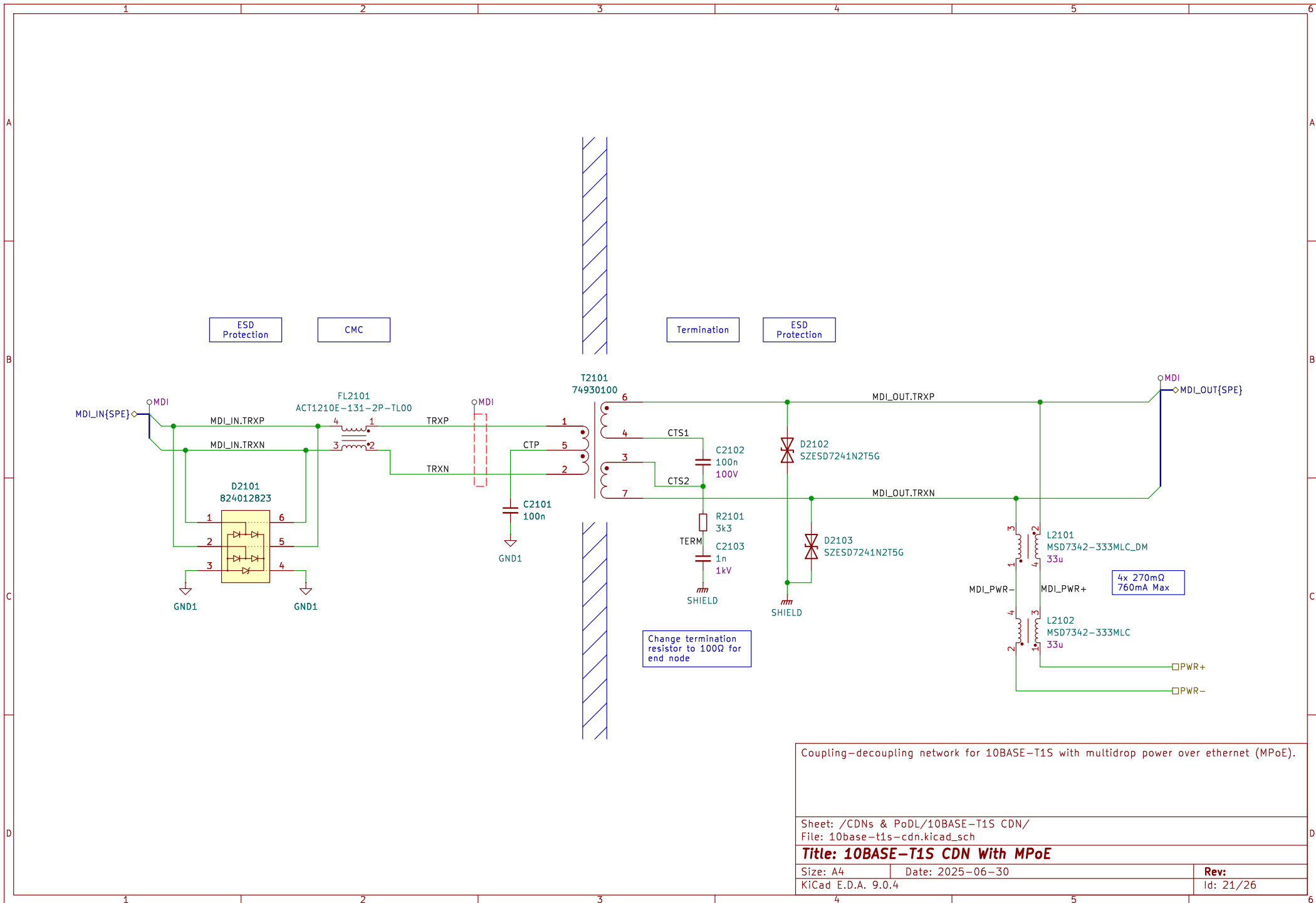
Title: 1000BASE-T1 CDN With PoDL

Size: A4 Date: 2025-06-30

KiCad E.D.A. 9.0.4

Rev:

Id: 20/26



Coupling-decoupling network for 10BASE-T1S with multidrop power over ethernet (MPoE).

Sheet: /CDNs & PoDL/10BASE-T1S CDN/
File: 10base-t1s-cdn.kicad_sch

Title: 10BASE-T1S CDN With MPoE

Size: A4

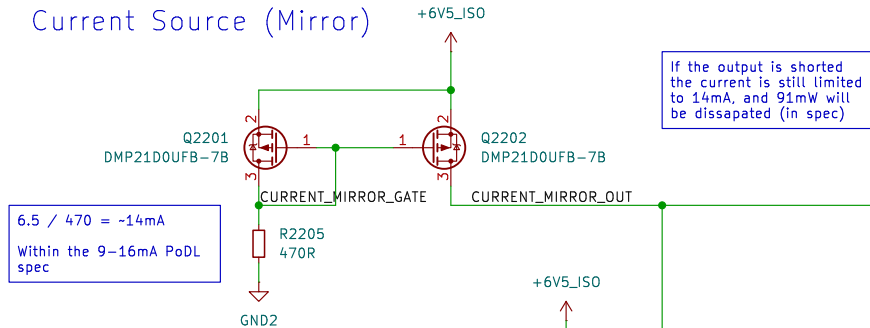
Date: 2025-06-30

Rev:

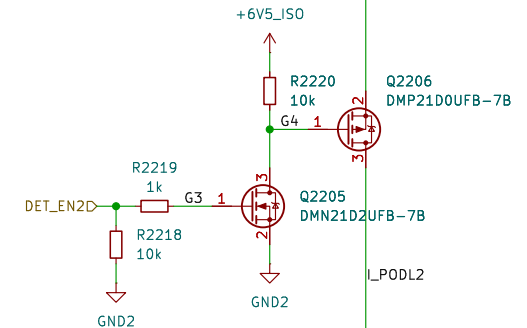
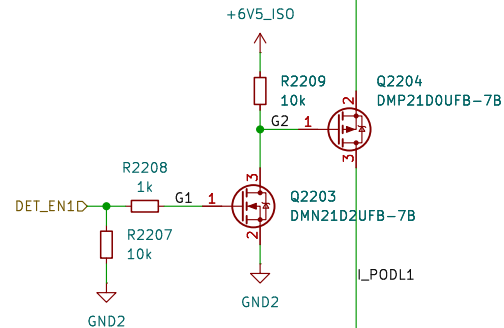
KiCad E.D.A. 9.0.4

Id: 21/26

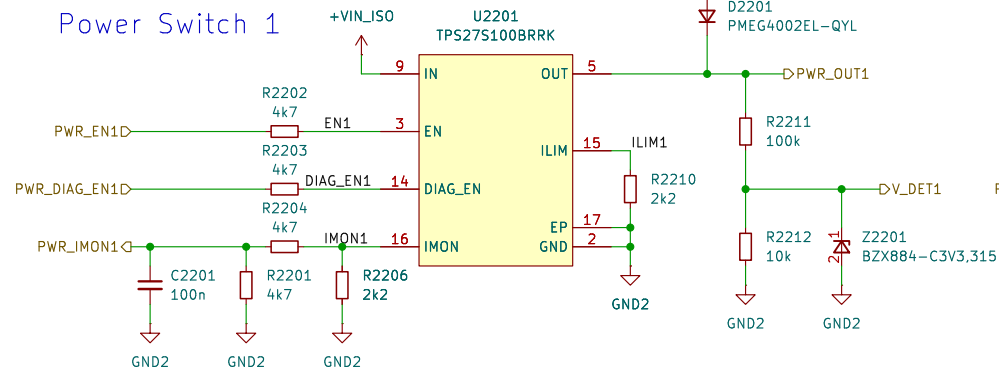
Current Source (Mirror)



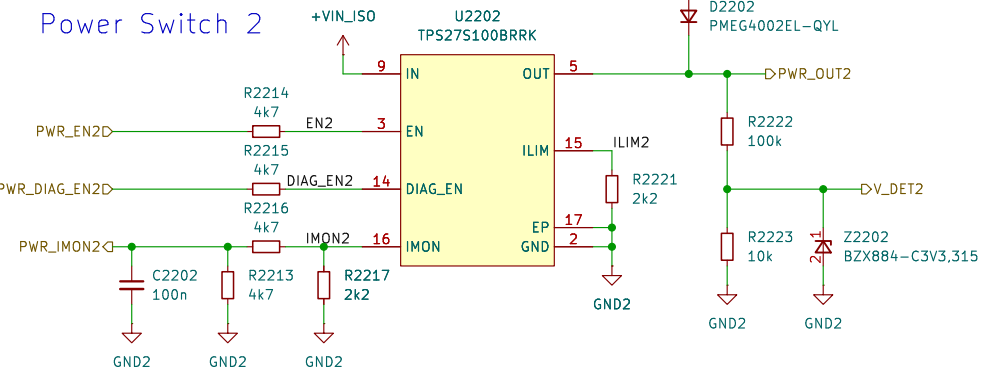
Current Switches



Power Switch 1



Power Switch 2



1.1A Maximum output current
1.1A Maximum measurable current

IMON can go up to 4.75V, potential divider limits this to be within 2.5V ADC range.

1/11 Potential divider and zener clamp to protect the 3V3 microcontroller IO.

$$I_{OUT} = 0.28046 \cdot V_{MON}$$

During the detection phase, 9-16mA is sent into TRXP and the resulting voltage is measured.
If a PoDL compatible device is detected on the other end (indicated by a 4.05-4.55V zener diode) then the full 24V is enabled.

Sheet: /CDNs & PoDL/PoDL (2 Channels)/
File: podl-2-channels.kicad_sch

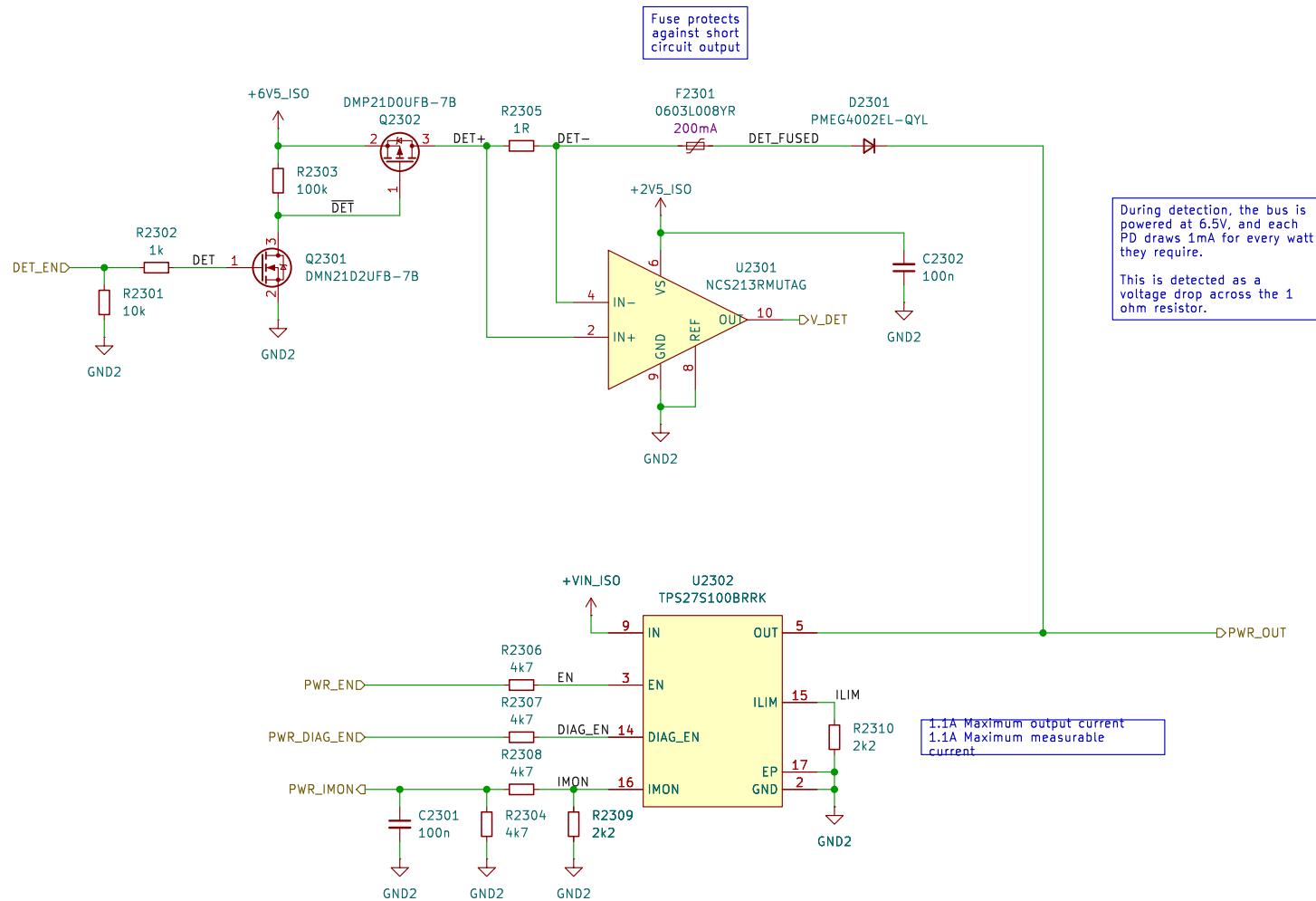
Title: PoDL Detection, Switching and Monitoring

Size: A4 Date: 2025-06-30

KiCad E.D.A. 9.0.4

Rev:

Id: 22/26



Sheet: /CDNs & PoDL/MPoE (1 Channel)/
File: mpoe-1-channel.kicad_sch

Title: MPoE Detection, Switching and Monitoring

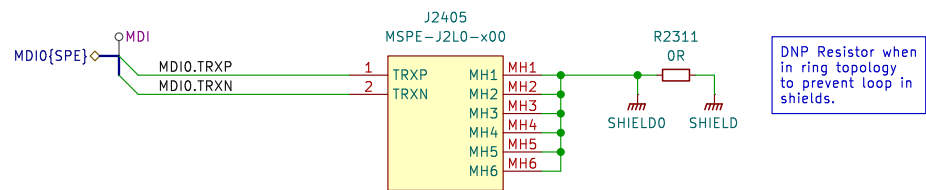
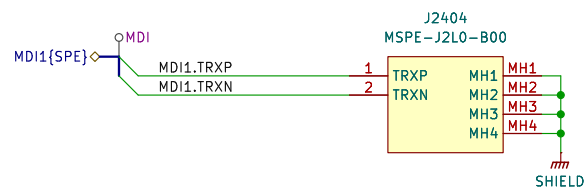
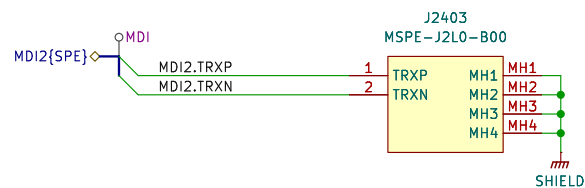
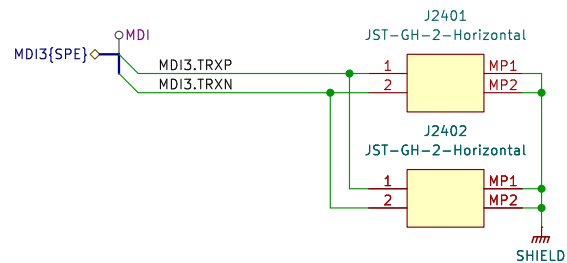
Size: A4

Date: 2025-06-30

Rev:

KiCad E.D.A. 9.0.4

Id: 23/26



Sheet: /Ethernet Connectors/
File: ethernet-connectors.kicad_sch

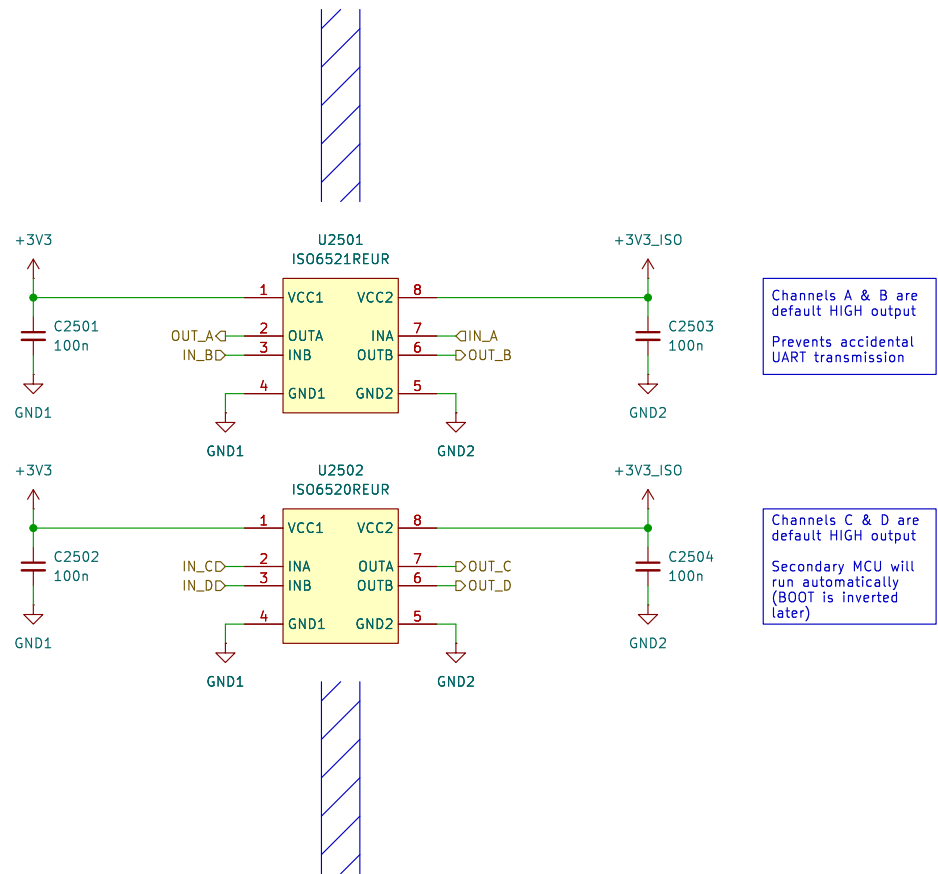
Title: Ethernet Connectors

Size: A4 Date: 2025-06-30

KiCad E.D.A. 9.0.4

Rev:

Id: 24/26



Sheet: /Isolation Crossing/		
File: isolation-crossing.kicad_sch		
Title: Isolation Crossing		
Size: A4	Date: 2025-06-30	Rev:
KiCad E.D.A. 9.0.4		Id: 25/26

